

Machine Learning A (2026)

Home Assignment 1

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Contents

1 Optimal Classification (16 points)	2
2 Logistic Regression (50 points)	2
2.1 Cross-entropy error measure (16 points)	2
2.1.1 Rewriting negative logarithmic likelihood as cross-entropy	2
2.1.2 Cross-entropy and logistic regression	3
2.2 Logistic Regression loss gradient (24 points)	3
2.2.1 Gradient for -1 and 1	3
2.2.2 Gradient for 0 and 1	3
2.2.3 Influence of misclassified examples	3
2.3 Log-odds	3
3 Sleep Well (34 points)	3
3.1 Data understanding and preprocessing	3

1 Optimal Classification (16 points)

1. From the lecture notes we know that if class posteriors are given the optimal classifier is $\hat{y} = \operatorname{argmax}_k (Y = C_k \mid X = x)$. We can define the function $g = \begin{cases} 1 & \text{if } x > 0 \\ 0 & \text{if } x \leq 0 \end{cases}$. In this case the optimal classifier is

$$\hat{y}(x) = g(P(Y = 1 \mid X = x) - P(Y = 0 \mid X = x))$$

We have $\hat{y}(0) = g(0 - 1) = 0$ and $\hat{y}(1) = g(0.2 - 0.8) = 0$, so

$$\hat{y}(x) = 0$$

And also $P(\text{error} \mid X = 1) = 1 - P(Y = \hat{y}(1) \mid X = 1) = 0.2$ and $P(\text{error} \mid X = 0) = 1 - P(Y = \hat{y}(0) \mid X = 0) = 0$. Since all inputs occur with equal probability $P(X = x) = 0.5$. The risk is therefore

$$R(\hat{y}) = \sum_{x \in \{0,1\}} P(X = x) * P(\text{error} \mid X = x) = 0.5 * (0.2 + 0) = 0.1$$

2. If $x = 0$ the probabilistic classifier is always correct, since it always correctly predicts the label $y = 0$, i.e. $P(\text{error} \mid X = 0) = 0$. For $x = 0$ if the true label is $y = 1$, which happens 80% of times, the classifier errors 20% of times. For $x = 1$ if the true label is $y = 1$, which happens 20% of times, the classifier errors 80% of times. So $P(\text{error} \mid X = 1) = 0.8 * 0.2 + 0.2 * 0.8 = 0.32$. The risk therefore is

$$R = 0.5(0 + 0.32) = 0.16$$

2 Logistic Regression (50 points)

2.1 Cross-entropy error measure (16 points)

2.1.1 Rewriting negative logarithmic likelihood as cross-entropy

We have that

$$h(x) = P(Y = 1 \mid X = x) = \frac{1}{1 + e^{-w^t x}} = \frac{e^{w^t x}}{1 + e^{w^t x}}$$

Thus

$$\begin{aligned} \mathcal{L}(w) &= \frac{1}{N} \sum_{n=1}^N \left[\mathbb{1}_{y_n=1} \ln \frac{1}{h(x_n)} + \mathbb{1}_{y_n=-1} \ln \frac{1}{1 - h(x_n)} \right] \\ &= \frac{1}{N} \sum_{n=1}^N \left[\mathbb{1}_{y_n=1} \ln(1 + e^{-w^t x_n}) + \mathbb{1}_{y_n=-1} \ln(1 + e^{w^t x_n}) \right] \\ &= \frac{1}{N} \sum_{n=1}^N \left[\ln(1 + e^{-y_n w^t x_n}) \right] \end{aligned}$$

2.1.2 Cross-entropy and logistic regression

2.2 Logistic Regression loss gradient (24 points)

2.2.1 Gradient for -1 and 1

We have

$$\begin{aligned}\nabla \mathcal{L}(w) &= \nabla \frac{1}{N} \sum_{n=1}^N \ln(1 + e^{-y_n w^t x_n}) \\ &= \frac{1}{N} \sum_{n=1}^N -\frac{1}{1 + e^{-y_n w^t x_n}} y_n x_n e^{-y_n w^t x_n} \\ &= \frac{1}{N} \sum_{n=1}^N -\frac{1}{1 + e^{y_n w^t x_n}} y_n x_n \\ &= \frac{1}{N} \sum_{n=1}^N -y_n x_n \theta(-y_n w^t x_n)\end{aligned}$$

2.2.2 Gradient for 0 and 1

2.2.3 Influence of misclassified examples

2.3 Log-odds

Let me define $p = P(Y = 1 | X = x)$. We have that σ as the logistic function satisfies the equation:

$$\begin{aligned}\sigma(w^T x + b) &= \sigma\left(\ln \frac{P(Y = 1 | X = x)}{P(Y = 0 | X = x)}\right) \\ &= \sigma\left(\ln \frac{p}{1-p}\right) \\ &= \frac{1}{1 + e^{-\ln \frac{p}{1-p}}} \\ &= \frac{1}{1 + \frac{1-p}{p}} \\ &= \frac{1}{\frac{p+(1-p)}{p}} \\ &= p = P(Y = 1 | X = x)\end{aligned}$$

3 Sleep Well (34 points)

3.1 Data understanding and preprocessing

The frequencies of the class labels was calculated using numpy: